# macOS Automated Device Enrollment with SSO (Jamf Pro + Intune)

Implementation Solution Guide

Version 1.0.0 · 2026-06-15

# Contents


Executive Summary — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 5

1. Solution Overview — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 5

What this solution is . . . 5

Why Jamf as the MDM (and Intune for identity & compliance) . . . 6

How it works . . . 6

Platform SSO authentication methods (choose one) . . . 7

Benefits . . . 7

Limitations (be honest with buyers) . . . 7

The standard deployment model . . . 8

2. Requirements — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 8

R1. Apple Business Manager (or Apple School Manager) account . . . 8

R2. Jamf Pro tenant with administrative access . . . 8

R3. Jamf Pro Apple Push Notification service (APNs) certificate . . . 8

R4. Automated Device Enrollment instance in Jamf Pro (ABM ↔ Jamf token) . . . 9

R5. Devices present in Apple Business Manager and assigned to Jamf . . . 9

R6. Jamf PreStage Enrollment for macOS . . . 9

R7. Supported macOS version . . . 9

R8. Company Portal app for macOS (deployed by Jamf) . . . 9

R9. Microsoft Enterprise SSO extension profile (deployed by Jamf) . . . 10

R10. Jamf Pro ↔ Microsoft Intune Device Compliance integration . . . 10

R11. Entra ID Conditional Access policy for macOS . . . 10

R12. Network access to Apple, Jamf, and Microsoft endpoints . . . 10

Global Requirements — Intune Solutions (category-level) 11

G1. Microsoft Entra ID tenant . . . 11

G2. Microsoft Intune subscription . . . 11

G3. Administrative roles . . . 11

G4. Device enrollment in Intune . . . 11

G5. Pilot security group . . . 11

G6. Supported, current operating system . . . 11

G7. Network access to required cloud endpoints . . . 12

3. Implementation — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 12

Phase 1 — Connect Apple to Jamf Pro . . . 12

Phase 2 — Assign devices and define the enrollment experience . . . 13
Phase 3 — Deploy Company Portal and Platform SSO (via Jamf) . . . 14
Phase 4 — Connect Jamf to Intune and enforce SSO . . . 14
Phase 5 — Pilot, validate, expand . . . 15
5. Testing & Validation — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 16
Test 1 — Jamf side readiness . . . 16
Test 2 — Intune/Entra side readiness . . . 16
Test 3 — Device appears as ADE in Jamf . . . 16
Test 4 — Zero-touch enrollment into Jamf . . . 17
Test 5 — Supervision and lock . . . 17
Test 6 — Company Portal + Platform SSO registration . . . 17
Test 7 — Single sign-on works . . . 17
Test 8 — Secure Enclave authentication method . . . 17
Test 9 — Compliance flows Jamf → Intune . . . 17
Test 10 — Conditional Access enforces SSO . . . 17
Acceptance gate . . . 18
6. Troubleshooting — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 18
The Mac never shows a "Remote Management" screen . . . 18
Enrollment fails partway / device not in Jamf . . . 18
Device enrolled but **not Supervised** . . . 18
Company Portal doesn't install . . . 18
Platform SSO never prompts to register . . . 19
Registered, but apps still prompt for a password . . . 19
Secure Enclave method not taking effect . . . 19
Mac doesn't appear in Intune / shows non-compliant . . . 19
Conditional Access blocks a Mac it shouldn't . . . 20
Integration renamed / steps don't match the console . . . 20
Token / certificate expiry (preventive) . . . 20
Useful checkpoints . . . 20
7. Best Practices — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 20
Architecture & ownership . . . 20
Identity & authentication . . . 21
Enrollment hygiene . . . 21
Token & certificate lifecycle (the #1 cause of outages) . . . 21

Operating system & updates . . . . . 21
Rollout discipline . . . . . 22
Operations & documentation . . . . . 22
Security posture . . . . . 22
8. Deliverables Summary — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune) 22
Documentation (this guide) . . . . . 22
PowerShell scripts — `JAMFADE-Scripts.zip` . . . . . 23
Hero image . . . . . 23
How it's delivered . . . . . 24
Appendix A — PowerShell Scripts 25

# Executive Summary — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

Apple **Automated Device Enrollment (ADE)** turns every company-owned Mac into a zero-touch, supervised, fully managed device the moment it is powered on — no imaging, no manual MDM steps, no local-admin hand-holding. This solution manages those Macs with **Jamf Pro**, the dedicated macOS management platform, and connects them to **Microsoft Entra ID** so sign-in is enforced with corporate identity and Conditional Access.

Most organizations that already run Jamf for their Mac fleet still need their Macs to be **first-class citizens in Microsoft Entra**: registered, compliant, and gated by the same Conditional Access policies as their Windows estate. This solution delivers exactly that, as a single, repeatable deployment a Mac admin can follow end to end:

- **Zero-touch provisioning with Jamf.** Macs in **Apple Business Manager** are assigned to a Jamf

Pro **Automated Device Enrollment** instance and a **PreStage Enrollment**, so they enroll supervised, branded, and pre-loaded the first time they run Setup Assistant.

- **Platform SSO with the Microsoft Enterprise SSO plug-in.** Jamf deploys the Single Sign-On

Extension configuration profile. Users sign in to macOS and to Microsoft 365 with their **Entra ID** identity, backed by a hardware-bound **Secure Enclave** key — phishing-resistant, not a synced password.

- **Compliance flows Jamf → Intune.** The **Jamf Pro ↔ Intune Device Compliance** integration

registers each Mac in Entra and reports its Jamf compliance state to Intune, so Macs appear in **Intune → Devices** as partner-managed and compliant.

- **SSO enforced by Conditional Access.** Entra **Conditional Access** requires a compliant,

registered Mac (and, optionally, a phishing-resistant authentication strength) before granting access to Microsoft 365 — turning Platform SSO from "available" into **mandatory**.

The package follows the standard repeatable shape — **Assess → Configure → Pilot → Validate → Expand** — with exact, click-by-click steps across both consoles, a requirements guide that tells you how to verify *and* fix each prerequisite, five ready-to-run PowerShell scripts (Microsoft Graph and the Jamf Pro API), and testing/troubleshooting guides with clear pass/fail gates.

**Outcome:** company Macs that Jamf provisions and manages, that authenticate with corporate identity via a phishing-resistant Secure Enclave key, and that Entra Conditional Access treats as compliant, registered, SSO-enforced devices — delivered by your own team, in a controlled pilot, without a consulting engagement.

# 1. Solution Overview — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

## What this solution is

This solution stands up company-owned Macs that **Jamf Pro manages** and **Microsoft Entra enforces**. It combines three capabilities into one repeatable deployment:

1. **Automated Device Enrollment (ADE)** — Apple's zero-touch enrollment program (formerly DEP).

Company-owned Macs that belong to your **Apple Business Manager (ABM)** organization are assigned to **Jamf Pro** as their MDM and enroll automatically — **supervised** — the first time they run Setup Assistant. Management is mandatory and locked to the device.

1. **Platform Single Sign-On (Platform SSO)** — Apple's macOS framework that lets a third-party

identity provider extend sign-in to the Mac itself. **Jamf deploys** the **Microsoft Enterprise SSO** extension via a configuration profile, so the Mac registers with **Microsoft Entra ID** and users authenticate with their corporate identity at the login window and across apps and Safari/Edge/Chrome.

1. **Jamf Pro ↔ Microsoft Intune Device Compliance** — the supported integration that registers each

Jamf-managed Mac in **Entra ID** and reports its **compliance** state to **Intune**. Combined with Entra **Conditional Access**, this is what makes SSO **enforced**: only a compliant, registered Mac is allowed into Microsoft 365.

Together they give you a Mac that Jamf provisions and manages, that signs in with corporate identity, and that Conditional Access treats as a trusted, compliant device.

## Why Jamf as the MDM (and Intune for identity & compliance)

A macOS device can have **only one MDM authority**. Organizations that have standardized on **Jamf Pro** for deep, Mac-native management (PreStage, policies, patch, Self Service) keep Jamf as the MDM — you do **not** move those Macs into Intune. Instead, you connect Jamf to the Microsoft cloud so the Macs become **Entra-registered and Intune-compliant**, and you enforce access with the **same Conditional Access** policies that protect the rest of your estate. Intune is the **identity and compliance authority**; Jamf is the **device management authority**. This is the only architecture in which "Jamf Pro device management" and "SSO enforced via Intune/Entra" coexist correctly.

## How it works

1. **ABM → Jamf Pro.** You connect ABM to Jamf Pro with an **Automated Device Enrollment** instance

(an ADE/MDM-server token). Macs assigned to the Jamf MDM server in ABM enroll into Jamf.

1. **PreStage Enrollment.** A **PreStage Enrollment** in Jamf defines the out-of-box experience:

supervised mode, account creation, which Setup Assistant panes to skip, and which apps/profiles pre-load.

1. **Power on → enroll.** When a user (or technician) powers on the Mac and connects to a network,

Setup Assistant contacts Apple, discovers the Jamf MDM server, and enrolls automatically. The device becomes supervised and managed by Jamf; your apps and profiles flow down.

1. **Platform SSO registration.** Jamf deploys a **Single Sign-On Extension** configuration profile

(Microsoft Enterprise SSO) and the **Company Portal** app. On first sign-in, Company Portal completes Platform SSO **registration**: the Mac generates a hardware-bound key in the **Secure Enclave** and registers with Entra ID.

1. **Compliance to Intune.** Through the **Jamf Pro ↔ Intune Device Compliance** connector, Jamf

sends the Mac's inventory and compliance state to Intune; the Mac appears in **Intune** → **Devices** as a partner-managed, compliant device.

1. **SSO enforced.** Entra **Conditional Access** requires a **compliant** (and optionally phishing-resistant) Mac before granting Microsoft 365 access. Platform SSO supplies the credential; Conditional Access makes it mandatory.

## Platform SSO authentication methods (choose one)

- **Secure Enclave key (recommended)** — a hardware-bound key in the Mac's Secure Enclave signs authentication. Phishing-resistant; the local password is decoupled from Entra. Best security.
- **Password sync** — the user's local macOS password is kept in sync with their Entra password. Simplest for users; least hardened.
- **Smart card** — authentication via a smart card/certificate. For high-assurance environments.

This solution standardizes on the **Secure Enclave key** method, with notes for the alternatives.

## Benefits

- **Zero-touch deployment** — Macs enroll supervised into Jamf on first boot; no imaging, no manual MDM steps.
- **Keep your Jamf investment** — full Mac-native management (PreStage, policies, patch, Self Service) stays in Jamf.
- **Corporate identity on the Mac** — Entra sign-in at the login window and across apps/browsers.
- **Phishing-resistant** — Secure Enclave key, hardware-bound, never leaves the device.
- **Compliance in Intune** — Macs appear as partner-managed, compliant devices alongside Windows.
- **SSO enforced** — Conditional Access blocks non-compliant or non-registered Macs from Microsoft 365.

## Limitations (be honest with buyers)

- A Mac has **one MDM**. In this design **Jamf is the MDM**; the Macs are **not** Intune-MDM-managed. Conditional Access works via the **partner compliance** integration, not Intune-native MDM.
- ADE requires devices to be in **Apple Business Manager** (purchased via Apple/an authorized reseller, or added with **Apple Configurator** — the latter adds a provisional 30-day release period).
- Platform SSO requires **macOS 13 Ventura or later** (macOS 14 Sonoma+ strongly recommended).
- Requires Jamf's **APNs (MDM Push) certificate** and the **ADE/ABM token**, both renewed annually — hard expiry dates that block enrollment if missed.
- Platform SSO registration and compliance reporting depend on the **Company Portal** app and the **Microsoft Enterprise SSO** extension being present and current.

- Microsoft and Jamf have **evolved** the integration over time (the legacy "Conditional Access" connector was superseded by **Device Compliance**). Always confirm the **current** integration name and steps in the Jamf and Microsoft documentation before connecting.
- Personally-owned (BYO) Macs use a different (user-enrollment) path and are out of scope here.

### The standard deployment model

This solution uses **one** model so the process is repeatable across our catalog:

> ***Company-owned, Apple Business Manager–sourced Macs, automatically enrolled into Jamf Pro (supervised), registered with Microsoft Entra ID and reporting compliance to Intune via the Jamf ↔ Intune Device Compliance integration, with Platform SSO via the Microsoft Enterprise SSO extension and a Secure Enclave key, and SSO enforced by Entra Conditional Access.***

Every step follows the repeatable pattern: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

## 2. Requirements — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.** Category-wide prerequisites (Entra tenant, Intune licensing, admin roles, pilot group, network) are covered once in the **Global Requirements — Intune Solutions** document; the items here are specific to Jamf-managed Apple ADE, Platform SSO, and the Jamf ↔ Intune Device Compliance integration.

### R1. Apple Business Manager (or Apple School Manager) account

**Description:** A verified Apple Business Manager (ABM) organization. **Why:** ADE only works for devices that belong to your ABM organization; ABM is also where you assign devices to the **Jamf Pro** MDM server and accept device-management terms. **Verify:** Sign in to `https://business.apple.com` with an ABM Administrator account; the organization status is **active**. **Implement:** Enroll at `https://business.apple.com/enroll` using your **D-U-N-S** number; complete Apple's verification. Allow 1–5 business days.

### R2. Jamf Pro tenant with administrative access

**Description:** A licensed **Jamf Pro** instance (cloud or on-prem) and an admin account with rights to manage enrollment, configuration profiles, and integrations. **Why:** Jamf Pro is the **MDM authority** in this design — it enrolls, supervises, and manages the Macs and deploys the SSO profile and Company Portal. **Verify:** Sign in to your Jamf Pro console (e.g. `https://yourorg.jamfcloud.com`) as an administrator; **Settings** is accessible. **Implement:** Provision Jamf Pro (Jamf Cloud is fastest) and assign an administrator role with **Settings**, **Enrollment**, and **Configuration Profiles** privileges.

### R3. Jamf Pro Apple Push Notification service (APNs) certificate

**Description:** A valid MDM Push certificate connecting **Jamf Pro** to Apple's push service. **Why:** Every Apple device managed by Jamf requires APNs; without it, no Mac can enroll or receive commands. (This

certificate lives in **Jamf**, not Intune.) **Verify:** Jamf Pro → **Settings** → **Global** → **Push certificates** (or **Global Management** → **Push Certificates**) shows a valid Apple **MDM Push** certificate with a future expiration. **Implement:** In Jamf Pro, create a Push Certificate request, sign in to `https://identity.apple.com` with a **company Apple ID** (not personal), upload the signed certificate. **Renew annually with the same Apple ID** — a new Apple ID re-issues the cert and unenrolls all devices.

## R4. Automated Device Enrollment instance in Jamf Pro (ABM ↔ Jamf token)

**Description:** An ABM **MDM server** that points to Jamf, with its server token (`.p7m`) uploaded into Jamf Pro as an **Automated Device Enrollment** instance. **Why:** This is the trust link that lets ABM-assigned devices enroll into your Jamf tenant. **Verify:** Jamf Pro → **Settings** → **Global** → **Automated Device Enrollment**; an instance is listed with a future token expiration. **Implement:** In Jamf Pro, download the **public key**; in ABM **Preferences** → **MDM Servers**, create an MDM server named clearly (e.g. **Jamf – Production**), upload the public key, download the server token; upload that token into Jamf's Automated Device Enrollment settings. **Renew annually.**

## R5. Devices present in Apple Business Manager and assigned to Jamf

**Description:** The target Macs appear in ABM and are assigned to the **Jamf** MDM server. **Why:** Only ABM devices can use ADE, and assignment tells Apple which MDM a device should enroll into during Setup Assistant. **Verify:** ABM → **Devices** → select the device → **MDM Server** shows your **Jamf** server. **Implement:** Purchase through Apple or an **Apple-authorized reseller** linked to your ABM (devices appear automatically), or add existing Macs with **Apple Configurator** (note: Configurator-added devices get a **30-day provisional period**). Then ABM → **Devices** → **Edit MDM Server** → choose the **Jamf** server, and in Jamf **refresh** the ADE instance.

## R6. Jamf PreStage Enrollment for macOS

**Description:** A macOS **PreStage Enrollment** in Jamf that defines the zero-touch experience. **Why:** The PreStage controls supervision, account creation, the Setup Assistant panes shown/hidden, and which configuration profiles and apps deploy at enrollment. **Verify:** Jamf Pro → **Computers** → **PreStage Enrollments**; a macOS PreStage exists and has devices scoped to it. **Implement:** Create a PreStage (Implementation Guide §Phase 2), enable **Supervision** and **Make MDM Profile Mandatory**, set account creation, and scope the target serial numbers.

## R7. Supported macOS version

**Description:** macOS **13 Ventura or later** (macOS **14 Sonoma+ recommended**). **Why:** Platform SSO and the Secure Enclave authentication method require macOS 13+; later releases materially improve registration reliability. **Verify:** Apple menu → **About This Mac**, or Jamf → device **Inventory** → **Operating System**. **Implement:** Upgrade via **System Settings** → **General** → **Software Update**, or enforce a minimum OS with a Jamf **Smart Group** + policy and a matching Intune **compliance** rule.

## R8. Company Portal app for macOS (deployed by Jamf)

**Description:** The Microsoft **Company Portal** application installed on the Mac by **Jamf**. **Why:** Company Portal performs the Platform SSO **registration** with Entra ID **and** the device registration that the Jamf ↔

Intune compliance integration depends on. **Verify:** Jamf Pro → **Computers** → **Configuration Profiles / Apps**; Company Portal is scoped to your pilot group and installs post-enrollment. **Implement:** Add the macOS Company Portal as a Jamf **package/app** (or via the App Installers / VPP) and scope it to the pilot Smart Group as **Required**.

### R9. Microsoft Enterprise SSO extension profile (deployed by Jamf)

**Description:** The **Microsoft Enterprise SSO** plug-in delivered via a Jamf macOS **Single Sign-On Extensions** configuration profile (Platform SSO). **Why:** This Apple-approved redirect extension brokers Entra authentication for Platform SSO and app/browser single sign-on. **Verify:** Jamf Pro → **Computers** → **Configuration Profiles**; a profile with a **Single Sign-On Extensions** payload (Platform SSO) exists targeting your pilot group. **Implement:** Create it during deployment (Implementation Guide §Phase 3), or run the included `JAMFADE-Set-JamfSSOProfile.ps1` to upload it via the Jamf Pro Classic API.

### R10. Jamf Pro ↔ Microsoft Intune Device Compliance integration

**Description:** The supported connector that registers Jamf-managed Macs in **Entra ID** and reports their **compliance** to **Intune**. **Why:** This is what makes the Macs appear as compliant in Intune and lets **Conditional Access** enforce SSO/compliance — without it, Jamf-managed Macs are invisible to Entra/Intune policy. **Verify:** Intune admin center → **Tenant administration** → **Connectors and tokens** → **Partner device management** (or **Partner compliance management**) shows **Jamf** as **Active**; in Jamf, **Settings** → **Global** → **Device Compliance** (macOS) is **connected**. **Implement:** In Intune, enable the **Partner device management / compliance** connection for Jamf; in Jamf, **Settings** → **Global** → **Device Compliance** → **macOS** → **Configure**, sign in with a Global Administrator to grant the cloud-connector app consent, and complete the connection. Confirm the exact current path in Jamf and Microsoft docs (the integration was renamed from the legacy "Conditional Access").

### R11. Entra ID Conditional Access policy for macOS

**Description:** A Conditional Access policy that requires a **compliant** (and optionally phishing-resistant) Mac for Microsoft 365. **Why:** This converts Platform SSO from "available" to **enforced** — non-compliant or non-registered Macs are blocked. **Verify:** Entra → **Protection** → **Conditional Access**; a policy targets **macOS** and requires **Require device to be marked as compliant** (start in **report-only**). **Implement:** Create the policy in Implementation Guide §Phase 4; pilot in **report-only**, then enable.

### R12. Network access to Apple, Jamf, and Microsoft endpoints

**Description:** Outbound HTTPS to Apple's enrollment/push/software endpoints, your Jamf Pro host, and Microsoft Entra/Intune endpoints. **Why:** Setup Assistant, ADE discovery, APNs, Jamf check-in, Entra registration, and compliance reporting all depend on these endpoints. **Verify:** From the device network, reach `https://mdmenrollment.apple.com`, `https://gdmf.apple.com`, the Apple push range `*.push.apple.com` (TCP 443 and 5223), your `*.jamfcloud.com` host, and `https://login.microsoftonline.com`. **Implement:** Allow Apple's published ranges (Apple support article HT210060), your Jamf Cloud host, and Microsoft's Entra/Intune ranges through proxy/firewall. Do **not** TLS-inspect Apple push/enrollment.

## Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level on the marketplace so individual solution listings stay small. Confirm these are in place before deploying *any* Intune solution.

Each entry: **Description · Why required · How to verify · How to implement if missing.**

## G1. Microsoft Entra ID tenant

**Description:** An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

## G2. Microsoft Intune subscription

**Description:** An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, enrollment, and security policies. **Verify:** `https://intune.microsoft.com` loads and **Tenant administration** → **Tenant status** shows an active subscription. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

## G3. Administrative roles

**Description:** Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies, enrollment connectors, and configuration profiles requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator** (or Global Administrator for connector setup). **Implement:** Assign **Intune Administrator** to the operator; reserve Global Administrator for tenant-wide changes.

## G4. Device enrollment in Intune

**Description:** Targeted devices are enrolled and manageable in Intune. **Why:** Policies only reach enrolled devices, regardless of platform (Windows, macOS, iOS/iPadOS, Android). **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Configure the appropriate enrollment method for the platform (e.g. automatic enrollment for Windows, Automated Device Enrollment for Apple) and enroll the device.

## G5. Pilot security group

**Description:** A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted to a small group before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists with intended members. **Implement:** Entra → **Groups** → **New group** → type **Security** → add pilot members.

## G6. Supported, current operating system

**Description:** A supported OS baseline with current updates for the targeted platform. **Why:** Modern Intune features and security settings assume a patched, vendor-supported OS. **Verify:** Check the device's OS version (e.g. `winver` on Windows, **About This Mac** on macOS) or Intune → device → **Hardware**. **Implement:** Deploy update policies (Windows Update rings, macOS update/compliance policies) or upgrade to a supported release.

### G7. Network access to required cloud endpoints

**Description:** Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs (plus Apple endpoints for Apple-platform solutions). **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges (and Apple's published ranges where applicable) through proxy/firewall.

**The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.**

# 3. Implementation — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

Follow these steps **in order**. The *why* for each capability lives in the Solution Overview; the steps below are deliberately short and exact so you can **click to implement**. Throughout, the repeatable pattern applies: **Assess → Configure → Pilot → Validate → Expand.** Do everything against a **pilot group / pilot Smart Group** first.

Consoles used:

- **Apple Business Manager** — `https://business.apple.com`
- **Jamf Pro** — your instance, e.g. `https://yourorg.jamfcloud.com`
- **Microsoft Intune admin center** — `https://intune.microsoft.com`
- **Microsoft Entra admin center** — `https://entra.microsoft.com`

> ***Integration naming note:*** *Microsoft and Jamf renamed the macOS integration from the legacy "Conditional Access" to* ***Device Compliance / Partner device management****. The steps below use the current model; confirm the exact menu labels in your tenant's Jamf and Microsoft docs.*

## Phase 1 — Connect Apple to Jamf Pro

### *Step 1. Create the Jamf APNs (MDM Push) certificate*

1. In Jamf Pro, go to **Settings** → **Global** → **Push certificates** (or **Global Management → Push Certificates**).
1. Click **New**, choose to create an **MDM Push Certificate**, and **download the CSR**.
2. Click through to **Apple Push Certificates Portal** (`https://identity.apple.com`) and sign in with a **company Apple ID** (record which Apple ID — you must reuse it every year).
1. **Create a Certificate**, upload the CSR, then **download** the signed `.pem`.
2. Back in Jamf, **upload** the `.pem` and enter the Apple ID used.
3. Confirm the certificate shows **valid** with an expiration ~1 year out.

### *Step 2. Create the Automated Device Enrollment instance (download Jamf public key)*

1. In Jamf Pro, go to **Settings** → **Global** → **Automated Device Enrollment**.
2. Click **New** and **download the public key** (`.pem`).

### *Step 3. Create the MDM server in Apple Business Manager*

1. In ABM, go to **Preferences (your name, bottom-left)** → **MDM Servers**.
2. Click **Add** (or **+**). Name it clearly, e.g. **Jamf – Production**.
3. Under **MDM Server Settings**, **Upload** the Jamf public key from Step 2.
4. Click **Save**, then **Download Token** — this downloads the server token (`.p7m`).

### *Step 4. Upload the ABM token into Jamf*

1. Back in Jamf's **Automated Device Enrollment** instance, **upload** the `.p7m` server token and enter the **Apple ID** used for ABM.

1. **Save.** The instance shows a future token expiration. Use **Refresh** to pull assigned devices.

## Phase 2 — Assign devices and define the enrollment experience

### *Step 5. Assign Macs to the Jamf MDM server (in ABM)*

1. In ABM, go to **Devices**.
2. Select the target Mac(s) by serial number (or filter by order).
3. Click **Edit MDM Server** → choose your **Jamf – Production** server → **Continue** → **Confirm**.

### *Step 6. Refresh and confirm devices in Jamf*

1. In Jamf Pro → **Settings** → **Global** → **Automated Device Enrollment** → **(your instance)**, click **Refresh** (Apple rate-limits syncs; allow a few minutes).

1. Confirm the assigned serial numbers appear under the instance.

### *Step 7. Create the macOS PreStage Enrollment*

1. In Jamf Pro, go to **Computers** → **PreStage Enrollments** → **New**.

2. **General:** name it (e.g. **macOS ADE – Pilot**); select the **Automated Device Enrollment instance** from Step 4.

1. Enable **Supervision** and **Make MDM Profile Mandatory** (prevents users from removing management).

1. **Account Settings:** configure the local account creation model (e.g. create a managed admin and a standard user, or defer user creation to Platform SSO per your design).

1. **Configuration Profiles / Enrollment Packages:** attach the **Single Sign-On Extensions** profile (Step 10) and the **Company Portal** app (Step 9) so they deploy at enrollment.

1. **Setup Assistant panes:** hide unnecessary panes (Restore, Apple ID, Siri, Analytics, Touch ID as policy dictates) for a clean first-run; keep **Location** and **Privacy** as required.

1. **Scope:** assign the pilot serial numbers. **Save.**

## Phase 3 — Deploy Company Portal and Platform SSO (via Jamf)

### *Step 8. Create the pilot Smart Group*

1. In Jamf Pro, go to **Computers** → **Smart Computer Groups** → **New**.

2. Name it **macOS SSO – Pilot**; add a criterion that matches your pilot devices (e.g. by PreStage, building, or serial list). **Save.**

### *Step 9. Add and scope the Company Portal app*

1. In Jamf Pro, add **Company Portal** for macOS — via **Computers** → **Mac Apps** (App Installers/VPP) or as a signed **package** under **Computers** → **Packages** + a policy.

1. Scope it to the **macOS SSO – Pilot** Smart Group, set to install automatically (Required).

### *Step 10. Create the Platform SSO configuration profile*

1. In Jamf Pro, go to **Computers** → **Configuration Profiles** → **New**.

2. **General:** name it **macOS Platform SSO – Pilot**; **Level = Computer Level**.

3. Add the **Single Sign-On Extensions** payload and configure:

• **Payload Type: SSO** (Redirect)

• **Extension Identifier:** `com.microsoft.CompanyPortalMac.ssoextension`

• **Team Identifier:** `UBF8T346G9`

• **Sign-On Type: Redirect**

• **URLs:** `https://login.microsoftonline.com`, `https://login.microsoft.com`, `https://sts.windows.net`, `https://login.partner.microsoftonline.cn`, `https://login.chinacloudapi.cn`, `https://login.microsoftonline.us`, `https://login-us.microsoftonline.com`

• **Platform SSO** → **Authentication Method: Secure Enclave Key** (recommended)

• Enable **Use Platform SSO** and (optional) **Token to user identifier** / new-user creation per your account model.

1. **Scope** to the **macOS SSO – Pilot** Smart Group. **Save.**

> *Prefer automation? Run the included* `JAMFADE-Set-JamfSSOProfile.ps1` *(Jamf Pro Classic API) to upload and scope this profile with the values above — then verify it in the Jamf console.*

## Phase 4 — Connect Jamf to Intune and enforce SSO

### *Step 11. Enable the Jamf ↔ Intune Device Compliance integration*

1. In **Intune** → **Tenant administration** → **Connectors and tokens** → **Partner device management** (or

**Partner compliance management**), enable the connection for **Jamf** (this consents the Jamf cloud-connector app in Entra).

1. In **Jamf Pro** → **Settings** → **Global** → **Device Compliance** → **macOS** → **Configure**.

2. Click **Connect**, sign in with a **Global Administrator** to grant consent, and complete the

connection. Confirm Jamf reports the connection as **active**.

1. In Intune, **Tenant administration** → **Partner device management** now shows **Jamf** as **Active**.

### *Step 12. Confirm device registration & compliance flow*

1. In Jamf, ensure the pilot Smart Group is **scoped for compliance** (the macOS Device Compliance

settings define which computers register and report).

1. After a pilot Mac registers (via Company Portal), confirm it appears in **Intune → Devices →

macOS **as** partner-managed **with a** compliance** state.

### *Step 13. Create the macOS Conditional Access policy (enforce SSO)*

1. In **Entra** → **Protection** → **Conditional Access** → **Create new policy**; **name it** `CA – macOS (Jamf) Compliant – Pilot`.

2. **Users:** your pilot group. **Target resources:** Office 365 (and other apps as needed).

3. **Conditions** → **Device platforms: macOS**.

4. **Grant: Require device to be marked as compliant** (optionally add **Require authentication

strength → Phishing-resistant** to require the Secure Enclave credential).

1. **Enable policy: Report-only** for the pilot. **Create.**

## Phase 5 — Pilot, validate, expand

### *Step 14. Pilot-enroll a test Mac*

1. **Erase** the pilot Mac (**System Settings → General → Transfer or Reset → Erase All Content and

Settings**) so it returns to Setup Assistant. *(For a Configurator-added device, complete this before the 30-day provisional period ends.)*

1. Power on, choose language/region, connect to network.

2. Setup Assistant shows a **Remote Management** screen naming your organization (Jamf).

3. Complete enrollment; the PreStage applies, **Company Portal** installs, and the **SSO** profile

deploys.

1. Open **Company Portal** and complete the **Register** prompt to finish Platform SSO registration

and Entra device registration (a key icon / login-window prompt confirms it).

### *Step 15. Validate*

1. Run `JAMFADE-JamfReadiness.ps1` (Jamf side: ADE instance, PreStage, SSO profile) and

`JAMFADE-Discovery.ps1` (Intune/Entra side: Jamf partner active, compliance policy, CA policy, Company Portal).

1. Run `JAMFADE-Validation.ps1` for the pilot Mac: confirm it is **registered** in Entra,

**compliant** in Intune (partner-managed), and the **SSO profile** is applied.

1. On the Mac, open a Microsoft 365 web app and confirm **silent SSO** (no password prompt).
2. Confirm the Conditional Access **report-only** result in Entra **sign-in logs** shows the policy

would apply and the device evaluates as **compliant**.

### *Step 16. Expand*

1. Once the pilot passes, widen the PreStage scope and the SSO/Company Portal/compliance Smart Group

to broader populations (or make the PreStage the default for the ADE instance).

1. Move the Conditional Access policy from **report-only** to **On** for macOS.
2. Roll out in waves, validating each wave with the scripts before widening.

# 5. Testing & Validation — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

Validate against a **pilot device** before expanding. Each test has an explicit **pass/fail gate**. Use the included PowerShell scripts for tenant-side checks (Microsoft Graph and the Jamf Pro API) and the on-device steps for the user experience.

## Test 1 — Jamf side readiness

**Procedure:** Run `JAMFADE-JamfReadiness.ps1` (Jamf Pro API). **Pass:** Jamf **APNs** valid (>30 days), an **Automated Device Enrollment** instance present with a token >30 days, at least one macOS **PreStage Enrollment**, and a **Single Sign-On Extensions** (Platform SSO) configuration profile present. **Fail:** Any missing or expiring within 30 days → resolve before enrolling devices.

## Test 2 — Intune/Entra side readiness

**Procedure:** Run `JAMFADE-Discovery.ps1` (Microsoft Graph). **Pass: Jamf** shows as an **Active** device-compliance partner for macOS, a macOS **compliance policy** exists, a **Conditional Access** policy targets macOS requiring compliance, and the macOS **Company Portal** is available. **Fail:** Partner not active, or compliance/CA policy missing → fix per Implementation §Phase 4.

## Test 3 — Device appears as ADE in Jamf

**Procedure:** In Jamf → Automated Device Enrollment → (instance), or run `JAMFADE-JamfReadiness.ps1`. **Pass:** The pilot serial is listed and **scoped to the PreStage**, ready for

enrollment. **Fail:** Serial missing → re-check ABM MDM-server assignment (R5) and **Refresh** the instance.

## Test 4 — Zero-touch enrollment into Jamf

**Procedure:** Erase the pilot Mac and run Setup Assistant on a managed network. **Pass:** A **Remote Management** screen names your organization (Jamf); enrollment completes without manual MDM-profile installation; the device appears in **Jamf** → **Computers** within minutes. **Fail:** No Remote Management screen → device not assigned/refreshed, or network blocks `mdmenrollment.apple.com` (R12).

## Test 5 — Supervision and lock

**Procedure:** Inspect the enrolled device in Jamf (or run `JAMFADE-JamfReadiness.ps1`). **Pass:** Device is **Supervised = Yes** and the MDM profile is **mandatory** (user cannot remove it). **Fail:** Supervised = No → the PreStage did not enable **Supervision** (Step 7); re-erase and re-enroll after correcting.

## Test 6 — Company Portal + Platform SSO registration

**Procedure:** After desktop loads, confirm Company Portal installed and complete the **Register** prompt; then check the SSO state. **Pass:** Company Portal reports the device **registered**; macOS shows the Platform SSO login-window prompt / key indicator; `JAMFADE-Validation.ps1` reports the device **registered** in Entra and the SSO profile **applied**. **Fail:** No registration prompt → confirm Company Portal is scoped/installed (R8) and the SSO profile is scoped (Step 10), then retry.

## Test 7 — Single sign-on works

**Procedure:** On the Mac, open a Microsoft 365 web app (e.g. `https://outlook.office.com`) in Safari and Edge/Chrome; open a native app that supports the broker. **Pass:** Sign-in is **silent** (no password prompt) using the registered Entra identity. **Fail:** Repeated password prompts → verify the SSO extension URLs and identifiers (Step 10), and that the Microsoft Enterprise SSO extension is present.

## Test 8 — Secure Enclave authentication method

**Procedure:** Confirm the Platform SSO method in effect. **Pass:** Authentication uses the **Secure Enclave key** (hardware-bound); the local password is decoupled from Entra. **Fail:** Method shows password sync when Secure Enclave was intended → correct the SSO profile (Step 10) and **re-register**.

## Test 9 — Compliance flows Jamf → Intune

**Procedure:** Review the pilot Mac in **Intune** → **Devices** → **macOS** (or run `JAMFADE-Validation.ps1` / `JAMFADE-DeviceReport.ps1`). **Pass:** The Mac appears as **partner-managed (Jamf)** with a **Compliant** state reported to Intune. **Fail:** Device absent or non-compliant → confirm the Device Compliance integration is **Active** (R10) and the Jamf compliance scope includes the device (Step 12).

## Test 10 — Conditional Access enforces SSO

**Procedure:** With the CA policy in **report-only**, sign in to Microsoft 365 from the Mac and review Entra **sign-in logs**. **Pass:** The CA policy evaluates the Mac as **compliant** and (if configured) satisfies the phishing-resistant strength; in **report-only** it shows "would have applied / granted." **Fail:** Policy reports

non-compliant or not-registered → resolve Tests 6/9 first, then re-test before switching CA to **On**.

### Acceptance gate

Proceed to **Expand** only when Tests 1–10 pass on at least one pilot Mac and the user-experience checks (silent SSO, clean first-run) are confirmed by a real pilot user. Switch Conditional Access from **report-only** to **On** only after Test 10 passes.

## 6. Troubleshooting — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

Work top-down: confirm the **Apple** ↔ **Jamf connection** (APNs/ADE token), then **assignment** (ABM → Jamf), then the **device** (enrollment/supervision), then **Platform SSO**, then the **Jamf** ↔ **Intune compliance** flow and **Conditional Access**. Most failures are an expired token, an unassigned device, a missing integration, or a blocked network endpoint.

### The Mac never shows a "Remote Management" screen

- **Device not assigned to the Jamf MDM server.** ABM → **Devices** → confirm **MDM Server** = your Jamf server (R5). After fixing, Jamf → Automated Device Enrollment → **Refresh**.
- **Not scoped to a PreStage.** Jamf → **PreStage Enrollments** → confirm the serial is scoped (R6).
- **Already past Setup Assistant.** ADE only triggers during Setup Assistant — **Erase All Content and Settings** and try again.
- **Network blocks Apple.** Ensure `mdmenrollment.apple.com` and `gdmf.apple.com` are reachable and **not** TLS-inspected (R12).
- **Configurator device past 30-day window** or released by the user — re-add and re-assign.

### Enrollment fails partway / device not in Jamf

- **Jamf APNs certificate invalid/expired.** Jamf → **Push certificates** must be valid (R3). If it was renewed with a *different* Apple ID, devices unenroll — restore the original Apple ID.
- **ADE token expired.** Renew the ABM server token (R4); re-upload to the Jamf ADE instance.

### Device enrolled but not Supervised

- The **PreStage** didn't enable **Supervision** (Step 7). Correct the PreStage, then **erase and re-enroll** — supervision is set at enrollment and cannot be toggled on an enrolled device.

### Company Portal doesn't install

- Confirm the app is scoped to the pilot Smart Group and set to install automatically (R8/Step 9).

• Force a check-in: on the Mac, run **Self Service** or `sudo jamf recon` / `sudo jamf policy`; in Jamf, confirm the computer's **inventory** updates.

## Platform SSO never prompts to register

• **Company Portal missing or outdated** — it performs registration (R8); update it.

• **SSO profile not scoped** to the device (Step 10). Confirm scope and that the profile installed (System Settings → Privacy & Security → Profiles).

• **Extension identifiers wrong** — must be `com.microsoft.CompanyPortalMac.ssoextension` / team `UBF8T346G9` (Step 10).

• **macOS too old** — Platform SSO requires macOS 13+ (14+ recommended) (R7).

• After fixing, sign out/in or reboot to re-trigger the registration prompt.

## Registered, but apps still prompt for a password

• **Missing SSO URLs** — ensure all Microsoft login URLs are listed in the SSO extension (Step 10).

• **Browser without broker support** — Safari/Edge/Chrome with the Microsoft SSO extension get silent SSO; some third-party browsers won't.

• **Conditional Access interruption** — a policy may require interaction; review CA sign-in logs in Entra.

## Secure Enclave method not taking effect

• The profile may be set to **password sync**. Set **Authentication Method = Secure Enclave Key** (Step 10), re-scope, and **re-register** (sign out/in). Existing registrations keep their original method until re-registered.

## Mac doesn't appear in Intune / shows non-compliant

• **Device Compliance integration not active.** Intune → **Tenant administration → Partner device management **must show** Jamf = Active **(R10). Re-run consent in Jamf** → Device Compliance** if needed.

• **Device not in the Jamf compliance scope.** Confirm the Smart Group used for compliance includes the Mac (Step 12).

• **Not registered in Entra.** Registration happens via **Company Portal** — complete Test 6 first.

• **Inventory stale.** Run `sudo jamf recon` to push fresh inventory; compliance reflects on the next sync.

## Conditional Access blocks a Mac it shouldn't

• **Compliance not yet reported.** New devices may take a sync cycle — confirm Intune shows **Compliant** before enforcing.

• **Policy too broad.** Keep CA in **report-only** for the pilot; review Entra **sign-in logs** to see exactly which control failed before switching to **On** (Step 13).

• **Authentication strength mismatch.** If you required phishing-resistant strength, the Mac must be using the **Secure Enclave** method (Test 8).

### Integration renamed / steps don't match the console

• Microsoft and Jamf **superseded the legacy "Conditional Access"** integration with **Device Compliance / Partner device management**. If a menu label differs, follow the** current** Jamf and Microsoft documentation for the macOS Device Compliance integration.

### Token / certificate expiry (preventive)

• The Jamf **APNs certificate** and the **ADE token** expire **annually** and **block enrollment** if lapsed. Run `JAMFADE-JamfReadiness.ps1` regularly; it flags anything expiring within 30 days. Renew APNs with the **same Apple ID** each time.

### Useful checkpoints

• **Jamf:** Computers → (device) → **Inventory / Management / Configuration Profiles**; Settings → Global → **Automated Device Enrollment / Push certificates / Device Compliance**.

• **Intune/Entra:** Devices → macOS → (device) → **Compliance**; Tenant administration → **Partner device management**; Entra** → Conditional Access **+** Sign-in logs**.

• **On the Mac: System Settings** → **Privacy & Security** → **Profiles**; **Console.app** filtered for `AppSSO` / `DeviceManagement`; `sudo jamf recon` to refresh inventory.

• **Scripts:** re-run `JAMFADE-Validation.ps1` after each change to confirm the gate.

## 7. Best Practices — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

Field-proven guidance to keep the Jamf-managed, Entra-enforced deployment reliable, secure, and low-maintenance.

### Architecture & ownership

• **Keep one MDM authority.** Jamf is the **MDM**; Intune/Entra are the **identity & compliance** authority. Never try to dual-enroll a Mac into both Jamf and Intune MDM.

• **Document the split.** Record which controls live in Jamf (enrollment, profiles, patch, Self

Service) versus Entra/Intune (registration, compliance, Conditional Access) so the team knows where to look.

## Identity & authentication

- **Standardize on the Secure Enclave key method.** It is phishing-resistant and decouples the local password from Entra. Reserve password sync for edge cases that genuinely need it.
- **Pair Platform SSO with a phishing-resistant Conditional Access strength** once the pilot is clean, so the Secure Enclave credential is actually required, not just available.
- **Enforce with Conditional Access in waves** — start **report-only**, confirm compliance is flowing, then turn the macOS policy **On**.

## Enrollment hygiene

- **Always supervise** company Macs (Step 7) and make the **MDM profile mandatory** so management can't be removed.
- **Buy through channels linked to ABM** so devices appear automatically; use Apple Configurator only for legacy stock, and complete enrollment before the **30-day provisional period** lapses.
- **Keep one well-named MDM server** per environment in ABM (e.g. *Jamf – Production*) to avoid assigning devices to the wrong tenant.
- **Use Smart Groups** to drive scope (PreStage, SSO profile, Company Portal, compliance) so a device picks up everything automatically as it enrolls.

## Token & certificate lifecycle (the #1 cause of outages)

- **Track both annual expiries** — the Jamf **APNs certificate** and the **ADE token**. A lapse blocks all new enrollments and can unenroll devices.
- **Always renew APNs with the same company Apple ID.** Document which Apple ID owns it; never use a personal Apple ID or one tied to an individual who may leave.
- **Schedule** `JAMFADE-JamfReadiness.ps1` to run regularly and alert on anything within 30 days of expiry.

## Operating system & updates

- **Set a minimum-OS rule in both places** — a Jamf Smart Group/policy and a matching Intune **compliance** rule (macOS 14+ recommended) so Platform SSO works reliably and out-of-support Macs are flagged and blocked by Conditional Access.
- **Manage updates in Jamf** (Software Updates / DDM) so the fleet stays patched without user

intervention.

### Rollout discipline

- **Pilot first, expand in waves.** Validate each wave with `JAMFADE-Validation.ps1` before widening.
- **Hide noisy Setup Assistant panes** for a clean, fast first-run, but keep privacy/location prompts that your policy requires.
- **Pre-stage apps and profiles** (Company Portal + SSO) in the PreStage so a freshly enrolled Mac is useful — and registerable — within minutes.

### Operations & documentation

- **Keep a runbook** of the Apple ID, ABM org, Jamf MDM server name, APNs/ADE expiry dates, the SSO profile identifiers, and the Device Compliance connector status — the values that, if lost, are painful to recover.
- **Report monthly** with `JAMFADE-DeviceReport.ps1`: macOS count, Platform SSO registration coverage, Entra registration, and Intune compliance (partner-managed) — so drift is visible early.
- **Re-validate the Jamf ↔ Intune integration after major updates** to either platform; Microsoft and Jamf occasionally change the connector and the SSO/Platform-SSO behavior between releases.

### Security posture

- **Combine with disk encryption** (FileVault, escrowed to Jamf) and a **macOS security baseline** for a complete managed posture.
- **Make compliance meaningful** — define Intune compliance to reflect real Jamf signals (encryption, OS version, etc.) so Conditional Access enforces a genuine security bar, not just registration.
- **Limit local admin** and use Jamf to manage privileged access; supervised + mandatory enrollment prevents users from quietly removing management.

## 8. Deliverables Summary — macOS Automated Device Enrollment with SSO (Jamf Pro + Microsoft Intune)

Everything included in this solution package, and what each piece is for.

### Documentation (this guide)

- **Executive Summary** — the business case and outcome in one page.

- **Solution Overview** — what ADE, Platform SSO, and the Jamf ↔ Intune Device Compliance integration

are, how they work together, why Jamf is the MDM and Intune the identity/compliance authority, the authentication methods, benefits, and honest limitations.

- **Requirements Guide** — every Apple/Jamf/Intune prerequisite with how to **verify** and how to **implement if missing**, plus the category-level Global Requirements.

- **Implementation Guide** — exact, ordered, click-by-click steps across five phases: connect Apple to Jamf, assign devices & define the PreStage, deploy Company Portal + Platform SSO via Jamf, connect Jamf to Intune and create the Conditional Access policy, then pilot/validate/expand.

- **Testing & Validation** — ten tests with explicit pass/fail gates and an acceptance gate.

- **Troubleshooting** — top-down diagnosis for enrollment, Platform SSO, compliance, and Conditional Access failures.

- **Best Practices** — architecture/ownership, identity, enrollment hygiene, the all-important token/certificate lifecycle, OS/updates, rollout discipline, and security posture.

## PowerShell scripts — `JAMFADE-Scripts.zip`

Each script has a header explaining **Purpose · How to run · Expected results**. Three use the **Microsoft Graph** PowerShell SDK (Intune/Entra side); two call the **Jamf Pro API** (MDM side).

1. `JAMFADE-Discovery.ps1` (Graph) — Intune/Entra readiness: the **Jamf device-compliance partner** is **Active** for macOS, a macOS **compliance policy** exists, a **Conditional Access** policy targets macOS, and the macOS **Company Portal** is available. Flags gaps.

1. `JAMFADE-Validation.ps1` (Graph) — validates a specific Mac: **Entra-registered**, reported by the Jamf partner, and **compliant** in Intune — with a clear PASS/FAIL summary.

1. `JAMFADE-DeviceReport.ps1` (Graph) — tenant-wide CSV report of macOS devices: registration, ownership, compliance state, and partner-managed (Jamf) status.

1. `JAMFADE-JamfReadiness.ps1` (Jamf Pro API) — Jamf-side readiness: APNs status/expiry, the **Automated Device Enrollment** instance/token expiry, macOS **PreStage** presence, and the **Single Sign-On Extensions** (Platform SSO) profile presence. Flags anything within 30 days of expiry.

1. `JAMFADE-Set-JamfSSOProfile.ps1` (Jamf Pro Classic API) — uploads and scopes the macOS Platform SSO (Single Sign-On Extensions) configuration profile with the correct Microsoft Enterprise SSO identifiers, URLs, and Secure Enclave method.

## Hero image

- `JAMFADE-hero.png` — the branded product image (1200×1200) used on the marketplace listing.

## How it's delivered

A branded **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

# Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

## JAMFADE-DeviceReport.ps1

**Purpose:** Produces a CSV of every macOS device known to Intune (including Jamf partner-managed records): device name, serial, OS version, ownership, management agent, compliance state, Entra registration, and last sync. Use it monthly to track Platform SSO registration coverage and Conditional-Access readiness across the Jamf-managed fleet.

**How to run:** Install-Module Microsoft.Graph -Scope CurrentUser # once .\JAMFADE-DeviceReport.ps1 # writes .\JAMFADE-macOS-report.csv .\JAMFADE-DeviceReport.ps1 -Path C:\Temp\macs.csv Reads only — makes no changes.

## JAMFADE-Discovery.ps1

**Purpose:** Confirms the Microsoft side is ready to treat Jamf-managed Macs as registered, compliant, Conditional-Access-gated devices: - the Jamf device-compliance PARTNER is Active for macOS, - at least one macOS COMPLIANCE policy exists, - a CONDITIONAL ACCESS policy targets macOS, - the macOS Company Portal app is available. This is the Intune/Entra counterpart to JAMFADE-JamfReadiness.ps1 (which checks the Jamf MDM side). Jamf — not Intune — is the MDM, so APNs/ADE tokens live in Jamf and are checked there.

**How to run:** Install the SDK once: Install-Module Microsoft.Graph -Scope CurrentUser Then: .\JAMFADE-Discovery.ps1 Reads only — makes no changes.

## JAMFADE-JamfReadiness.ps1

**Purpose:** Confirms the Jamf Pro side is ready to auto-enroll Macs and deploy Platform SSO: - an Automated Device Enrollment (ADE) instance with a token >30 days from expiry, - at least one macOS PreStage Enrollment, - a configuration profile containing the Single Sign-On Extensions (Platform SSO) payload. This is the Jamf counterpart to JAMFADE-Discovery.ps1 (which checks the Intune/Entra side). Jamf — not Intune — holds the APNs and ADE tokens, so they are validated here.

**How to run:** # Use a Jamf Pro account (or API client) with read access to enrollment + config profiles. $cred = Get-Credential # Jamf username / password (or API client id/secret) .\JAMFADE-JamfReadiness.ps1 -JamfUrl https://yourorg.jamfcloud.com -Credential $cred # Or pass an existing bearer token: .\JAMFADE-JamfReadiness.ps1 -JamfUrl https://yourorg.jamfcloud.com -BearerToken <token> Reads only — makes no changes.

## JAMFADE-Set-JamfSSOProfile.ps1

**Purpose:** Uploads a macOS Single Sign-On Extensions (Platform SSO) configuration profile to Jamf Pro with the correct Microsoft Enterprise SSO identifiers, login URLs, and the Secure Enclave authentication method — the same profile described in the Implementation Guide §Phase 3 / Step 10 — and scopes it to a target Smart Computer Group. Jamf is the MDM, so the SSO profile is delivered by Jamf (not Intune).

**How to run:** $cred = Get-Credential # Jamf account with config-profile write # Preview only (no change): .\JAMFADE-Set-JamfSSOProfile.ps1 -JamfUrl https://yourorg.jamfcloud.com -Credential $cred -TargetGroup "macOS SSO - Pilot" -WhatIf # Create & scope: .\JAMFADE-Set-JamfSSOProfile.ps1

-JamfUrl https://yourorg.jamfcloud.com -Credential $cred -TargetGroup "macOS SSO - Pilot" # Choose a different method: .\JAMFADE-Set-JamfSSOProfile.ps1 -JamfUrl ... -Credential $cred -TargetGroup ... -AuthMethod passwordSync

## JAMFADE-Validation.ps1

**Purpose:** Confirms that one specific Mac is, on the Microsoft side: - REGISTERED in Microsoft Entra ID (a macOS/MacMDM device object exists), - reported by the Jamf partner and present in Intune as a managed device, - COMPLIANT (the state Jamf reports through the Device Compliance integration). Use after a pilot Mac completes Company Portal registration (Implementation Test 6/9).

**How to run:** Install-Module Microsoft.Graph -Scope CurrentUser # once .\JAMFADE-Validation.ps1 -SerialNumber C02XXXXXXXXX # or match by device name: .\JAMFADE-Validation.ps1 -DeviceName "Jane-MacBook" Reads only — makes no changes.